Faculty of Computing

[Computer Communications & Network]

Lab No 6 Tasks

NAME: SITARA REHMAN

SAP: 66454

<u>Task 1:</u> Write the IP address 222.1.1.20 mask 255.255.255.192 in CIDR.

<u>Solution:</u>

The IP address 222.1.1.20 having mask 255.255.255.192 /24 24+2=26

So 255.255.255.192/26 because it has 26 bits to set 1 in binary.

<u>Task 2:</u> Write is the IP address 135.1.1.25 mask 255.255. 248.0 in CIDR notation.

<u>Solution:</u>

The IP address 135.1.1.25 having mask 255.255.248.0/21

So 255.255.248.0/21

Because it has 21 bits to set 1.

<u>Task 3:</u> You have been allocated a class C network address of 201.1.1.0 how may hosts can you have?

<u>Solution:</u>

So the subnet mask will be 255.255.255.0 now we have to find the number of hosts .

2^32-2=256-2=254.

Task 4: You have been allocated a class A network address of 21.0.0.0. You need create at least 10 networks and each network will support a maximum of 100 hosts. Would the following two subnet masks Work.

255.255.0.0 and or 255.255.255.0

Solution:

We have to create at least the 10 networks and 100 hosts.

Firstly 21.0.0.0 and subnet masks are

Mask: 255.255.0.0 (/16)

- Subnets from Class A (/8 → /16): 2^(16−8) = **256 networks** (≥10).
- Hosts per subnet: 2^(32−16) − 2 = **65,534 hosts** (≫100; functionally meets “up to 100,” but very wasteful).

Mask: 255.255.255.0 (/24)

- Subnets from /8 → /24: 2^(24−8) = **65,536 networks** (≥10).
- Hosts per subnet: 2^(32−24) − 2 = **254 hosts** (≥100 ;).

Task 5: You have been allocated a Class B network address of 129.1.0.0. You have subnetted it using the subnet mask 255.255.255.0 How many networks can you Have and how many hosts can you place on each network?

Solution:

Class B default is /16; given mask is **/24**.

- **Number of subnets:** 2^ (24−16) = **256 networks**.
- **Hosts per subnet:** 2^(32−24) − 2 = **254 hosts**.

256 subnets, each with 254 usable hosts.